



MME 321 Lab 11: Control System Design

Kyle Buttray

Dr. Hasan

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1.0 Open Loop Control Definition and Schematic

An Open Loop DC motor was modeled during the course of this experiment. An Open Loop Control system is characterized by control input being applied directly to the system. In this case an input of ___ of angular velocity is applied to the system, without feedback or reference.

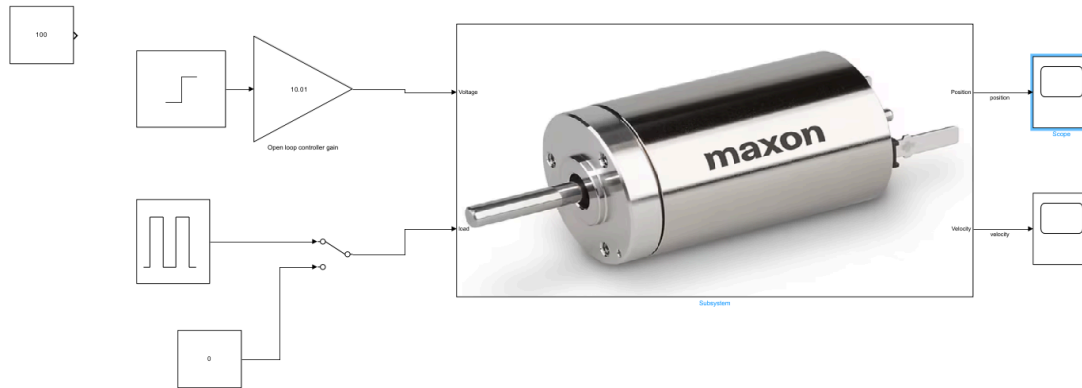


Fig. 1 Open Loop DC motor simulation schematic

2.0 Closed Loop Control Definition and Schematic

Two Closed Loop DC motors were also modeled during this experiment. A Closed Loop Control system notably uses feedback to obtain a closer approximation of a determined reference point. In this specific lab, a PID controller is simulated to create the closed loop control of the DC motor.

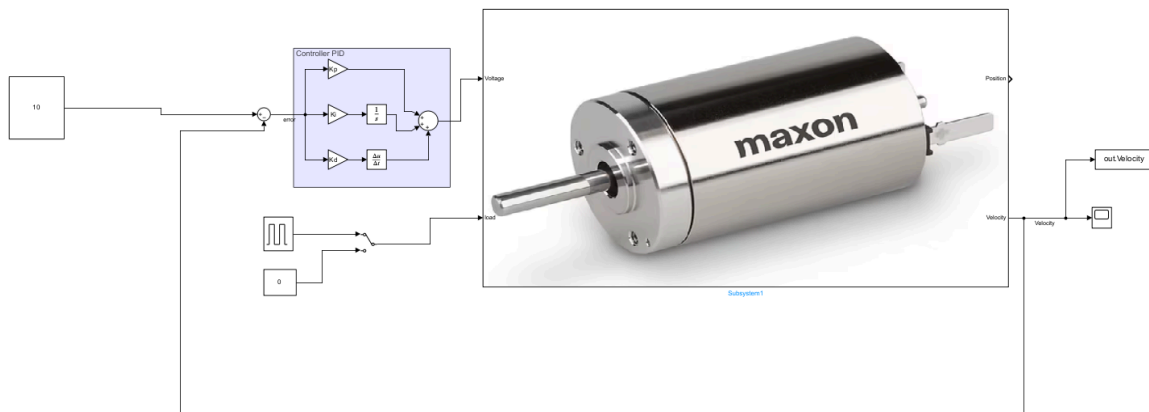


Fig. 2 Closed Loop DC motor simulation schematic

3.0 Transfer Function Derivation

The Transfer function from input voltage to angular velocity or angular position can be determined using the Laplace transform. However, this method assumes that all initial conditions are equal to 0.

As determined in previous labs, the electrical to mechanical transfer function is,

$$(Js + b)\Omega(s) = K_t I_a(s) - T_L$$

Solving for $I_a(s)$ and substituting into the electrical equation in laplace domain gives:

$$V_a(s) = (L_a s + R_a) \frac{(Js+b)\Omega(s)}{K_t} + K_e \Omega(s)$$

Therefore, the transfer voltage to angular velocity transfer function can be written as,

$$G(s) = \frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + b) + K_t K_e}$$

Additionally, since the angular position is simply the integral of angular velocity, we can write the voltage to angular position transfer function as follows:

$$G(s) = \frac{\theta(s)}{V_a(s)} = \frac{K_t}{s((L_a s + R_a)(Js + b) + K_t K_e)}$$

4.0 State-Space Model Derivation

For Voltage-to-Velocity State-Space Model, States are

$$x_1 = i_a, x_2 = \omega$$

Input is,

$$u = [V_a; T_L]$$

Output is

$$y = \omega$$

And written as

$$\dot{x}_1 = -(R_a/L_a) x_1 - (K_e/L_a) x_2 + (1/L_a) V_a$$

$$\dot{x}_2 = (K_t/J) x_1 - (b/J) x_2 - (1/J) T_L$$

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

$$x = [x_1; x_2], u = [Va; TL]$$

$$A = [(-Ra/La), (-Ke/La); (Kt/J), (-b/J)]$$

$$B = [(1/La), 0; 0, (-1/J)]$$

$$C = [0, 1]$$

$$D = [0, 0]$$

Voltage-to-Position State-Space Model

$$x_1 = \theta, x_2 = \omega, x_3 = ia$$

$$u = [Va; TL]$$

$$y = \theta$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = (-b/J)x_2 + (Kt/J)x_3 - (1/J)TL$$

$$\dot{x}_3 = (-Ke/La)x_2 - (Ra/La)x_3 + (Va/La)$$

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

$$x = [x_1; x_2; x_3], u = [Va; TL]$$

$$A = [0, 1, 0; 0, (-b/J), (Kt/J); 0, (-Ke/La), (-Ra/La)]$$

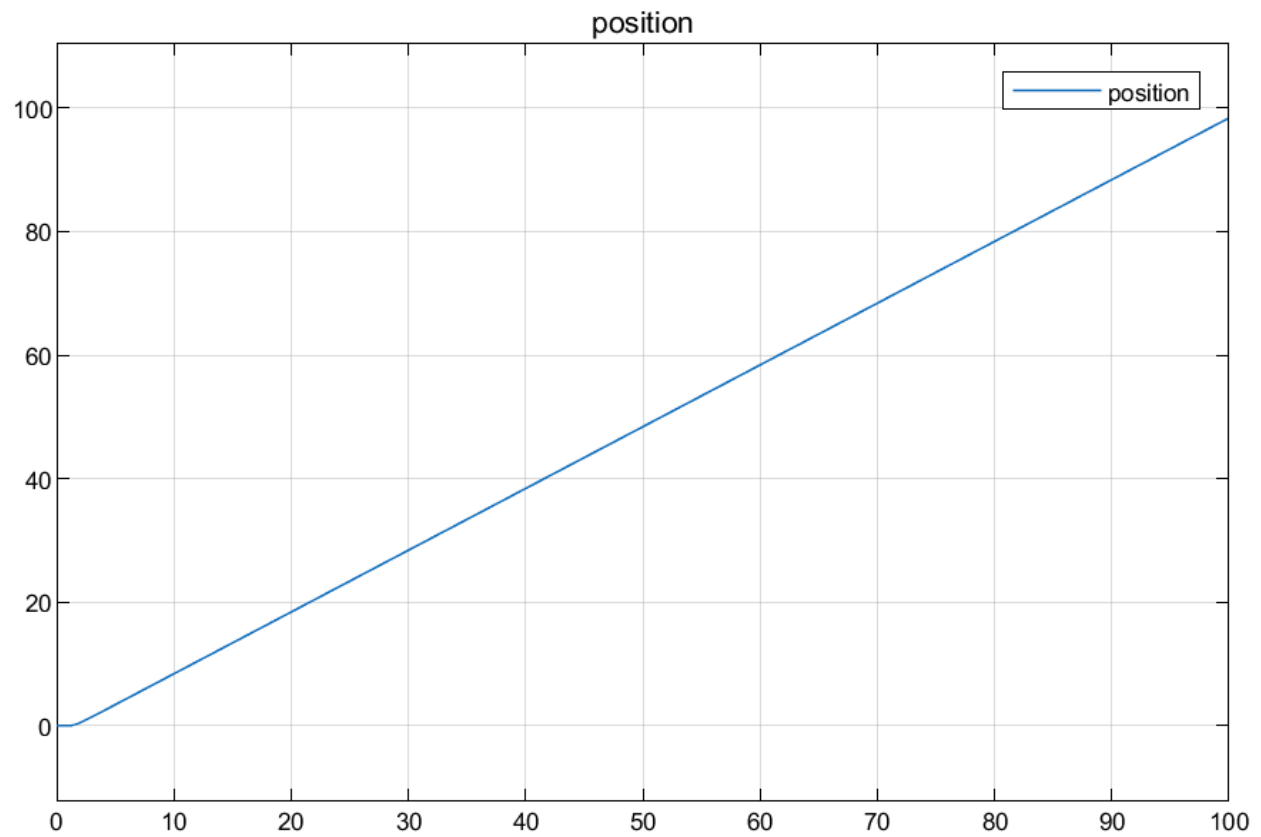
$$B = [0, 0, 0; (-1/J), (1/La), 0]$$

$$C = [1, 0, 0]$$

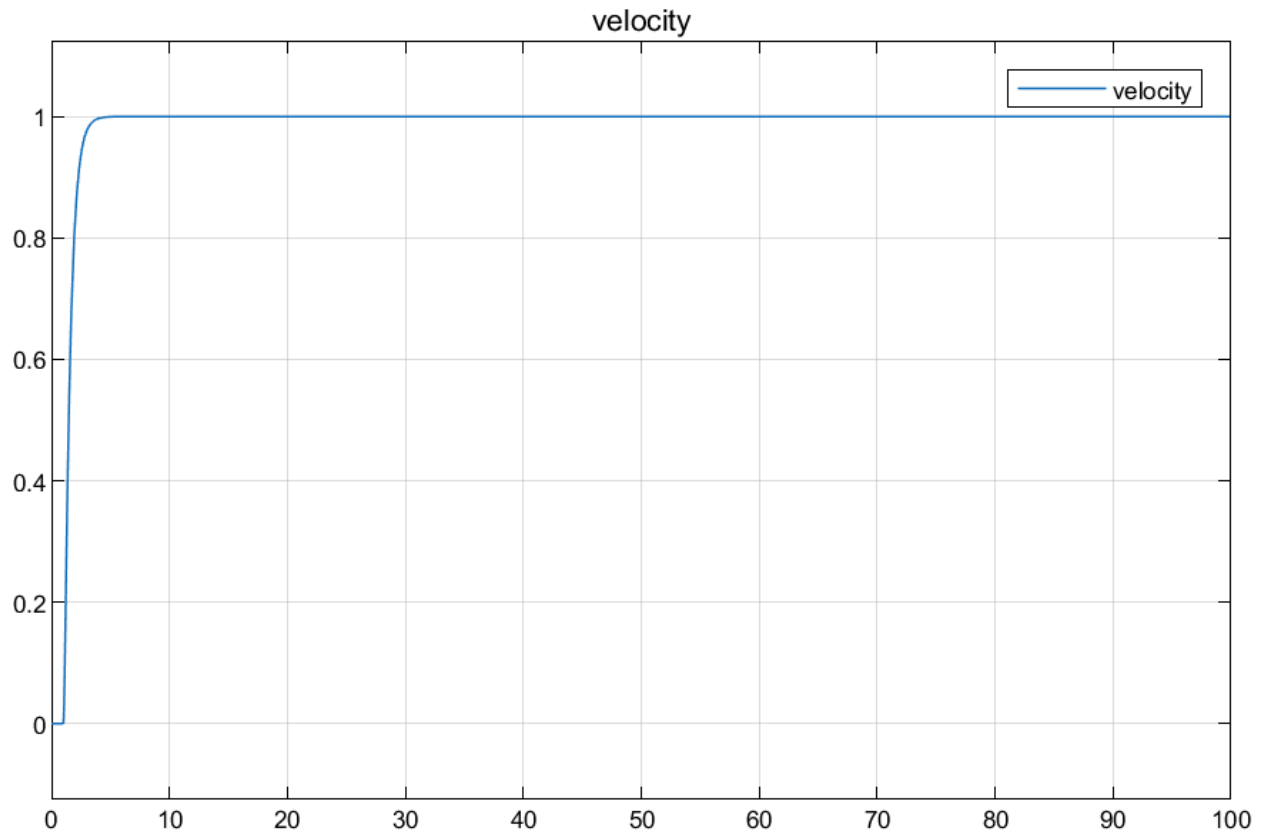
$$D = [0, 0]$$

5.0 Motor Model Step Response

It is difficult to see without zooming in, but the step response acts as a switch, instantly changing from a 0 to a 1. In place of the input speed block, it is set to immediately activate the motor with an input of 1 rad/s starting after 1 second. This can be seen in the position and velocity plots of the step motor as shown in plot. 1 and plot. 2.



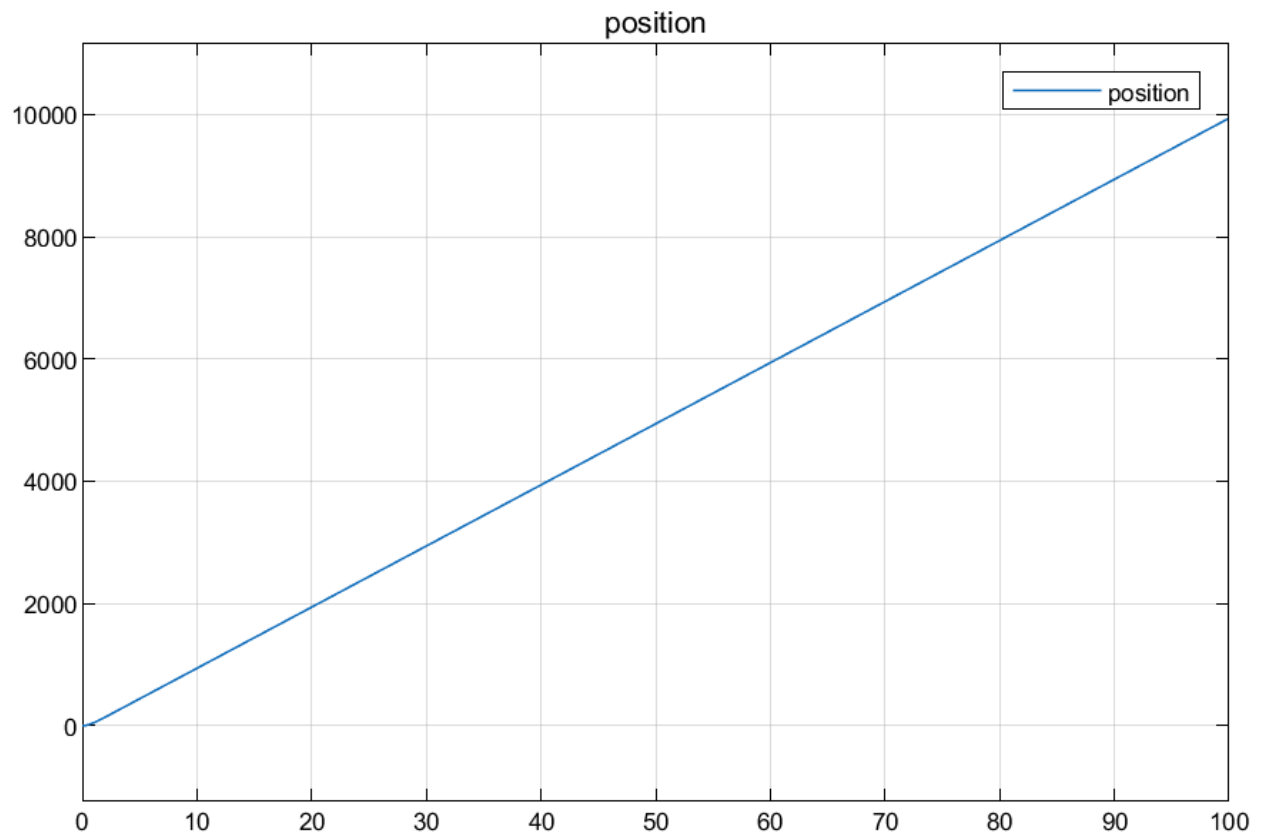
Plot. 1 Open Loop Motor Position as a step response (No Disturbance)



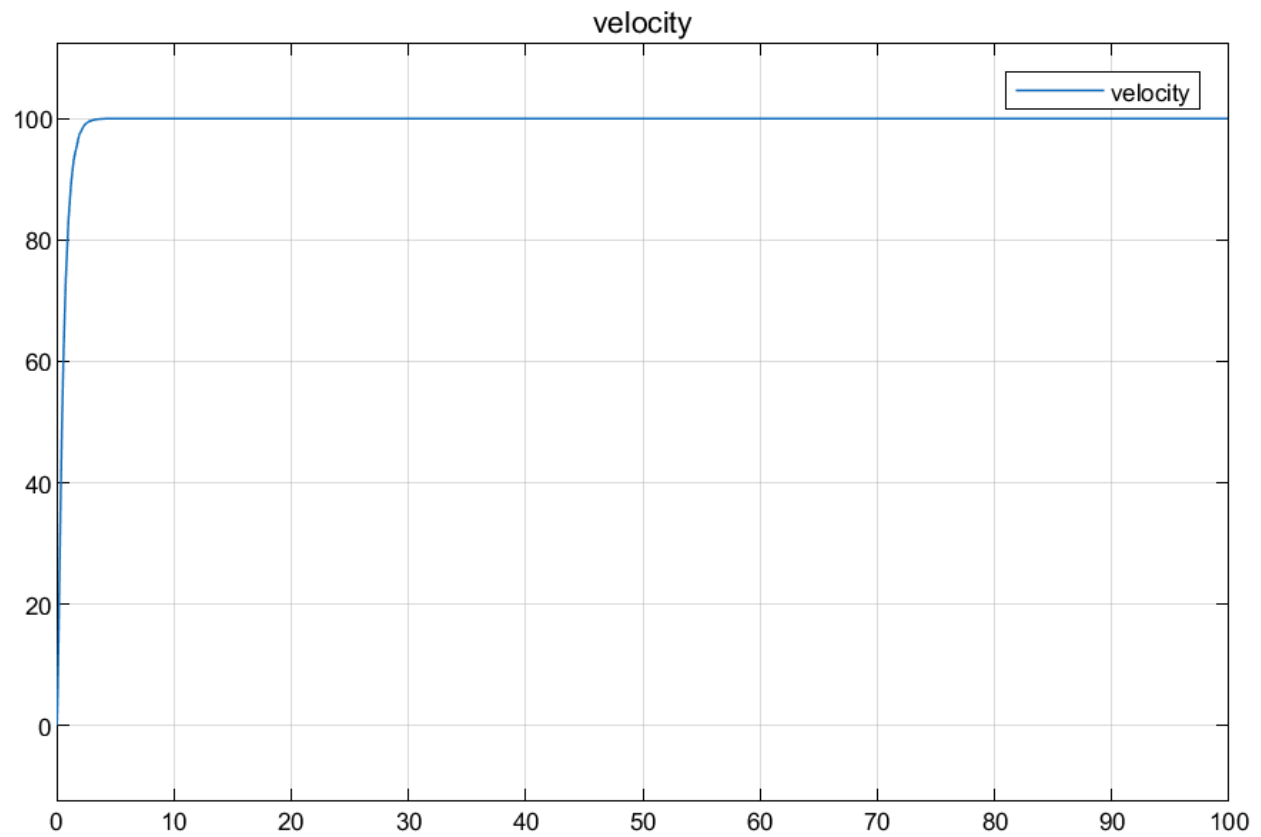
Plot. 2 Open Loop Motor Velocity as a step response (No Disturbance)

6.0 Open Loop Response

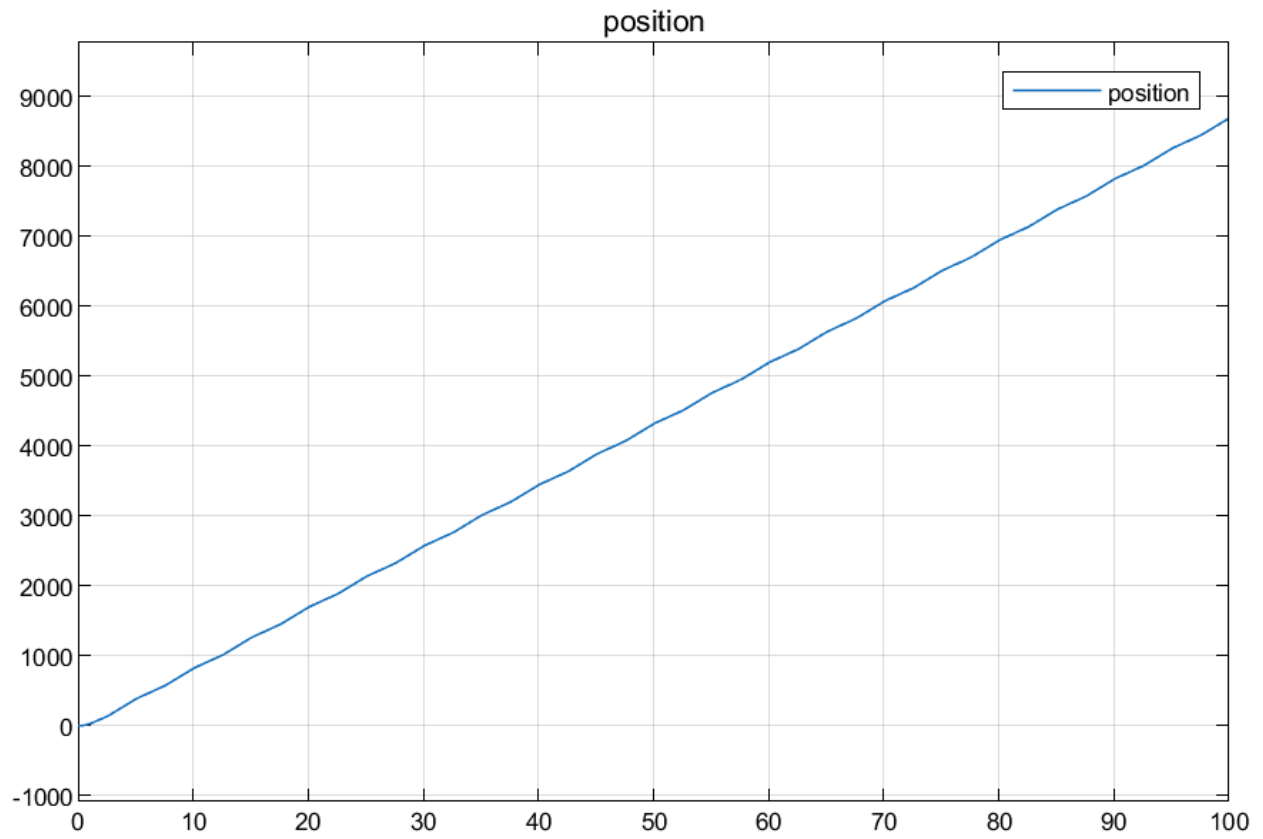
The open loop motor does not have a feedback loop and so therefore, once it reaches its target output, it does not change. This is extremely easy to see in the plots with undisturbed responses, as there is no overshoot when the response reaches steady state. However, this also extremely volatile overshoot within the disturbed simulations. This is because the motor does not know how to compensate for disturbance once there is overshoot in open loop control.



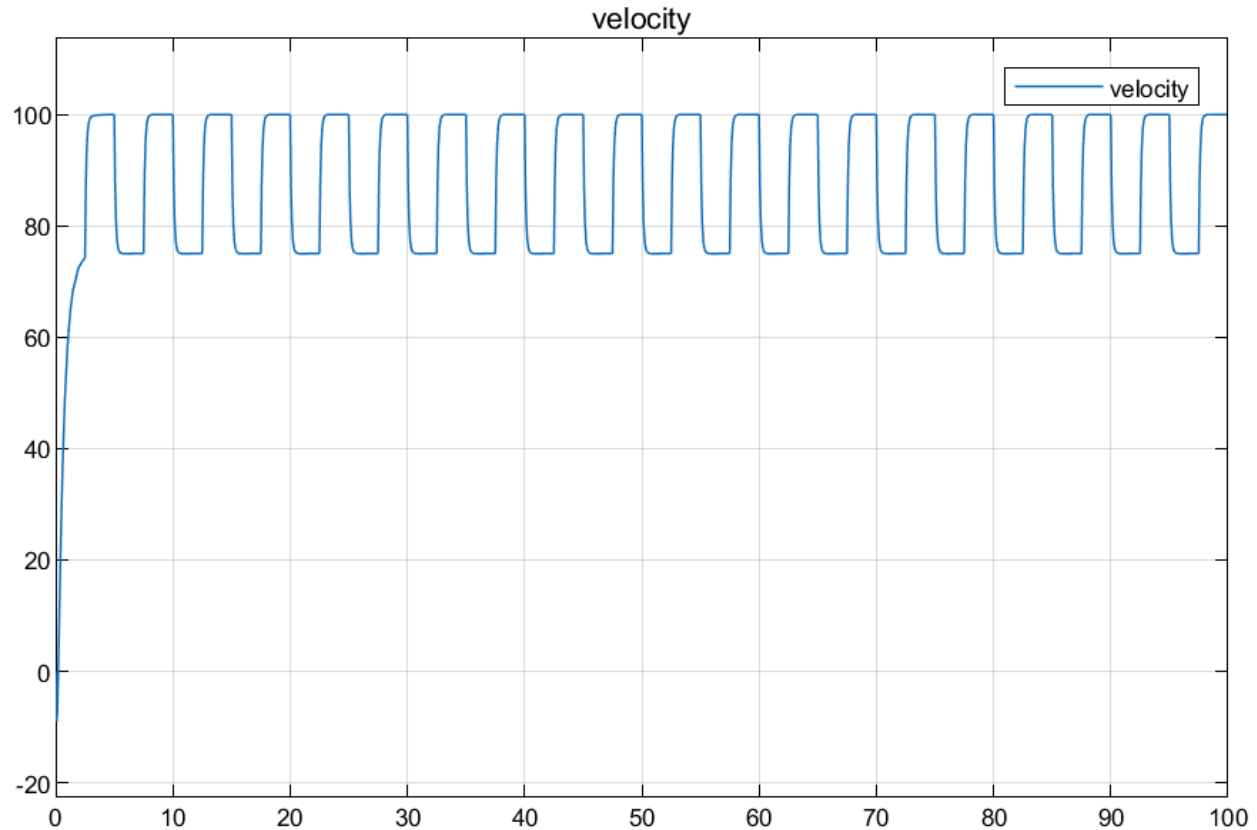
Plot. 3 Open Loop Motor Position as a function of time (No Disturbance)



Plot. 4 Open Loop Motor Velocity as a function of time (No Disturbance)



Plot. 5 Open Loop Motor Position as a function of time (with Disturbance)



Plot. 6 Open Loop Motor Velocity as a function of time (With Disturbance)

7.0 Closed Loop Response

Two motors of different input and output were examined for the closed loop responses. Unlike the open loop responses, even in ideal conditions the DC motor has some overshoot once it reaches steady state. However, due to the PID feedback controller the overshoot is much smaller and much more manageable until it is dampened out. That being said, it seems as though even the closed loop systems still are struggling with disturbances as shown in the plots below.

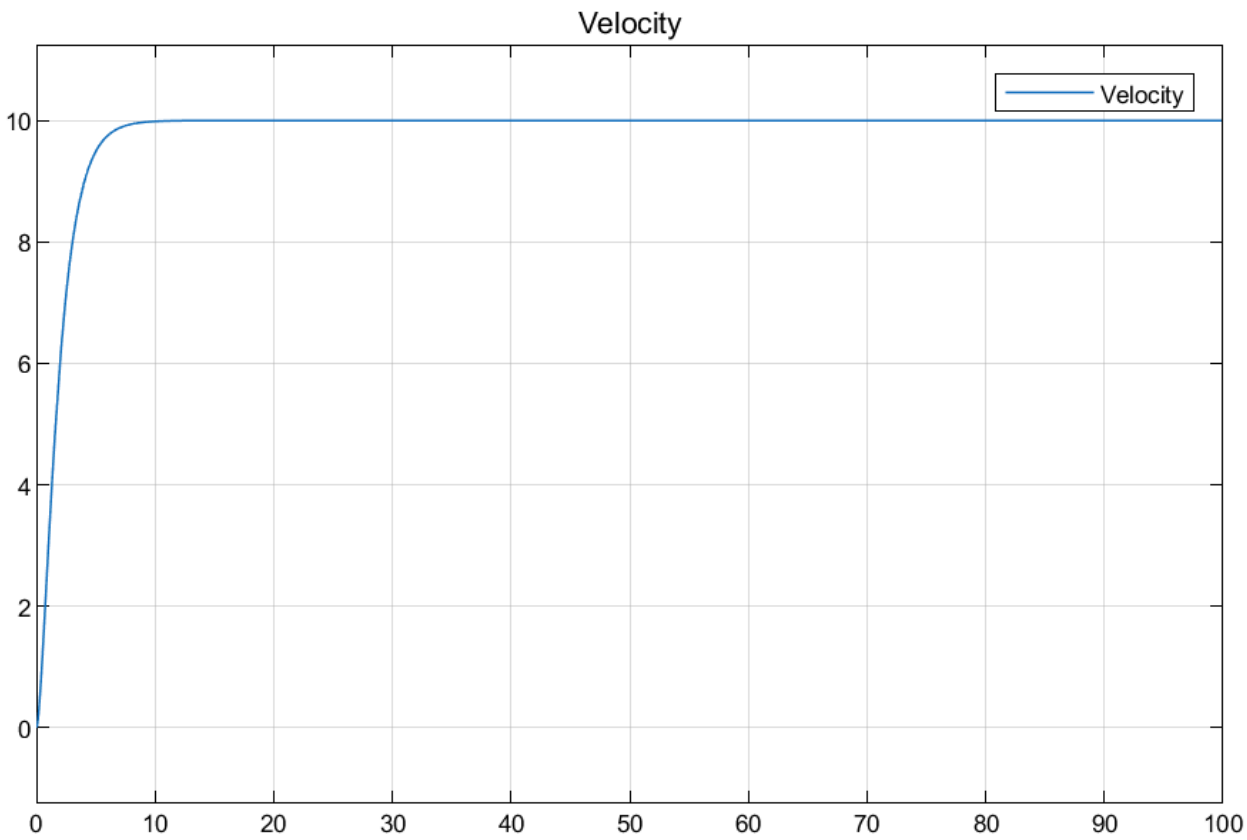
Input of 10 rad/s

Parameters:

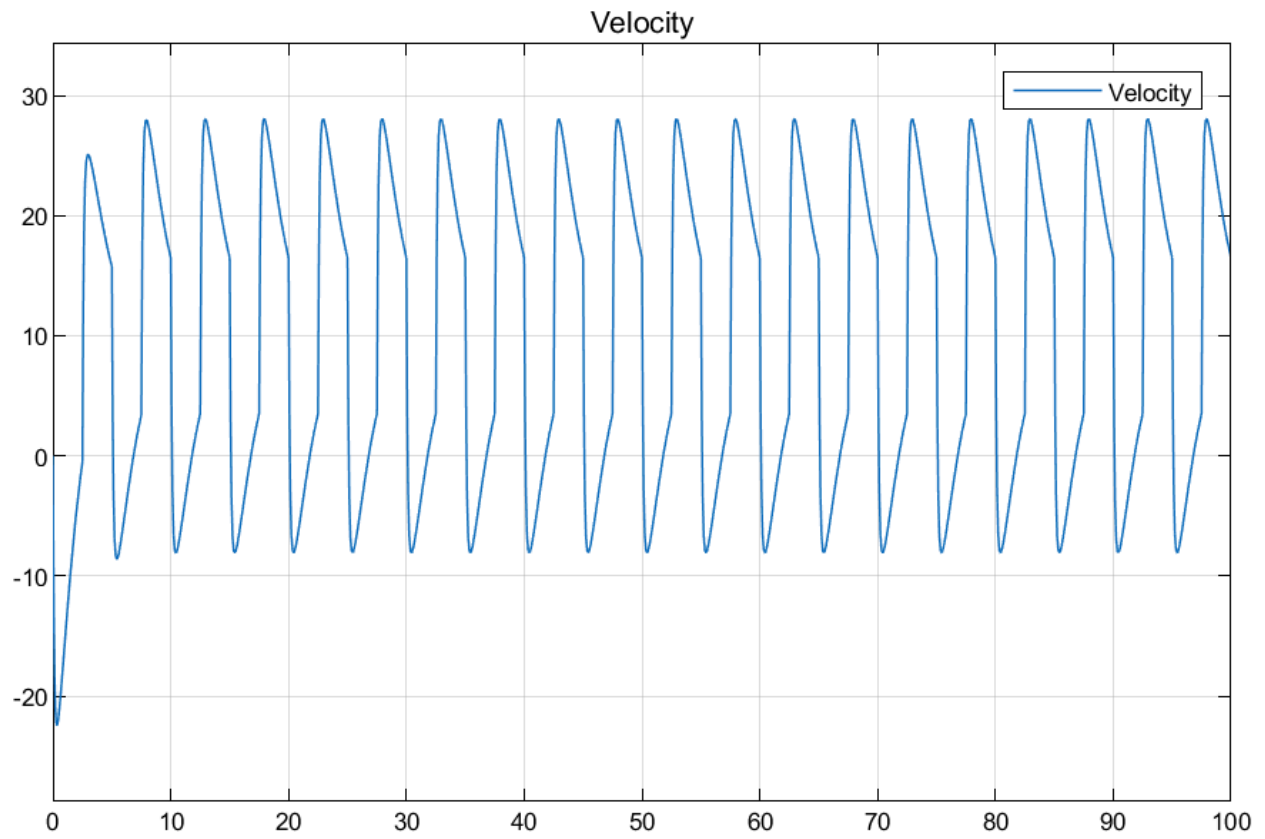
Set	Kp	Ki	Kd
V1	1.0	5.0	0.1

V2	3.0	12.0	0.03
V3	6.0	20.0	0.08

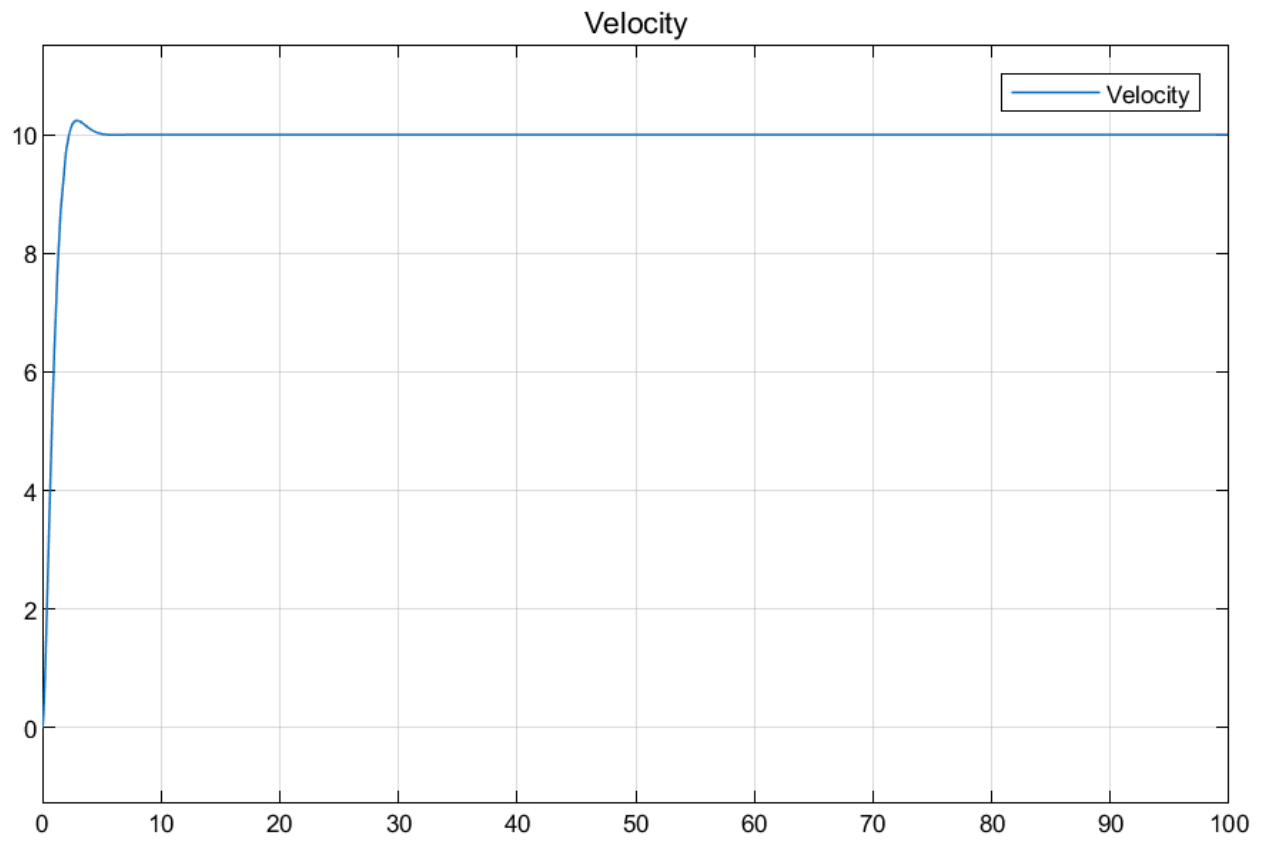
Table. 1 Input Parameters for closed loop motor of 10 rad/s input



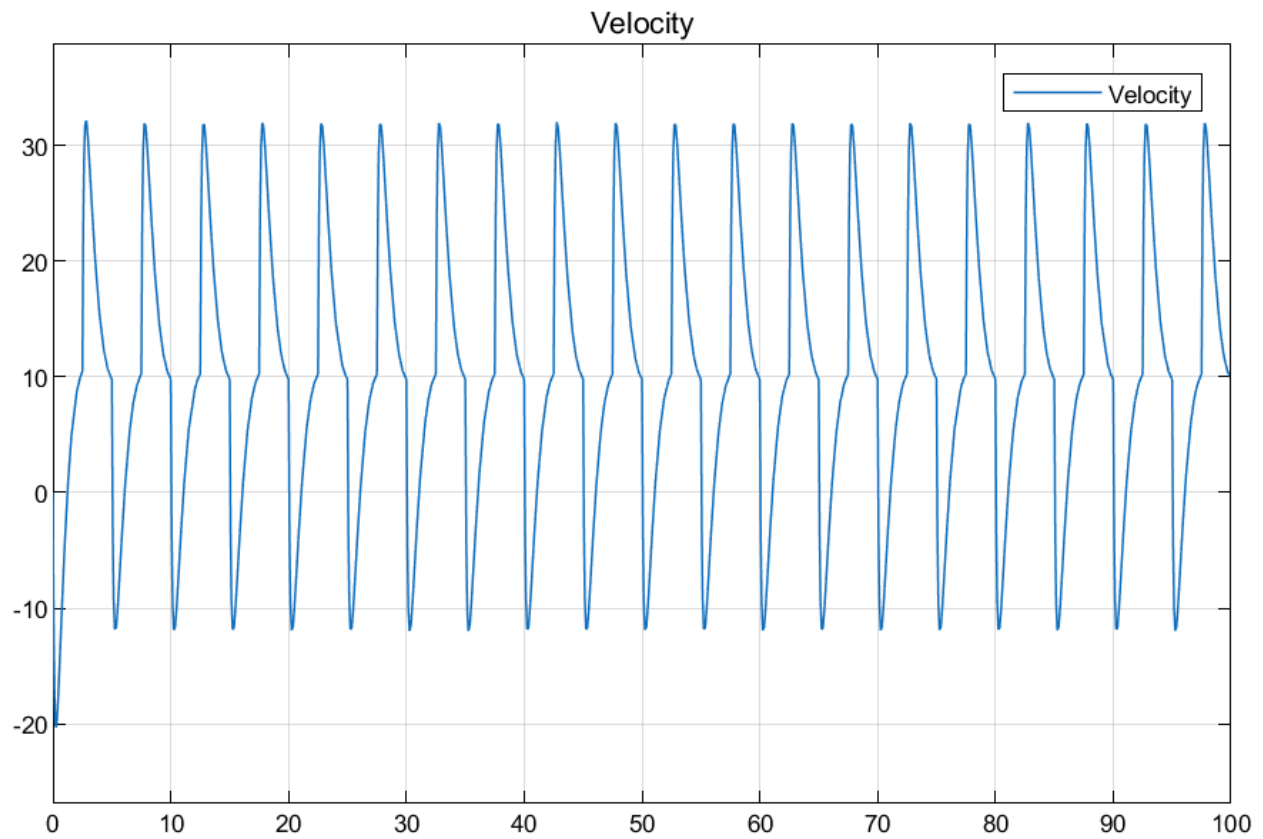
Plot. 7 [V1] Closed Loop Motor Velocity as a function of time (No Disturbance)



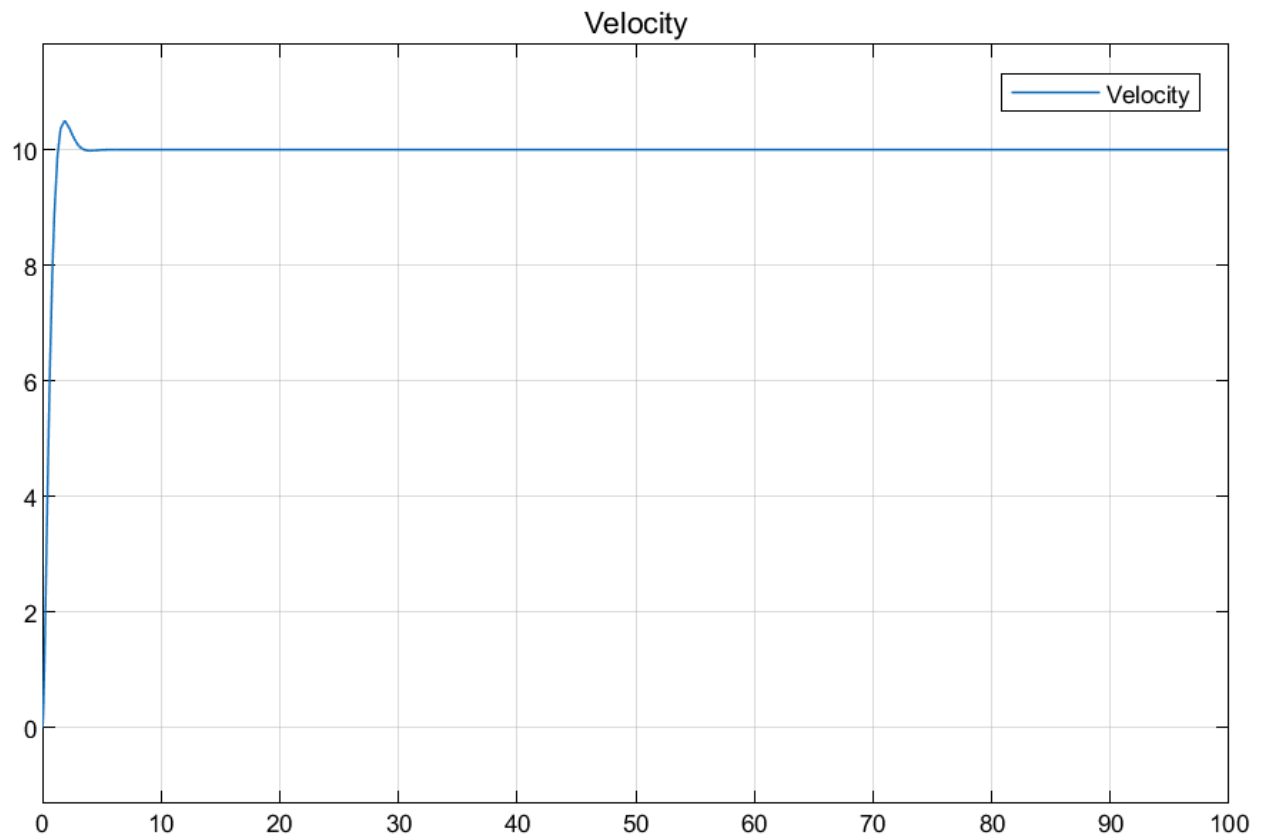
Plot. 8 [V1] Closed Loop Motor Velocity as a function of time (With Disturbance)



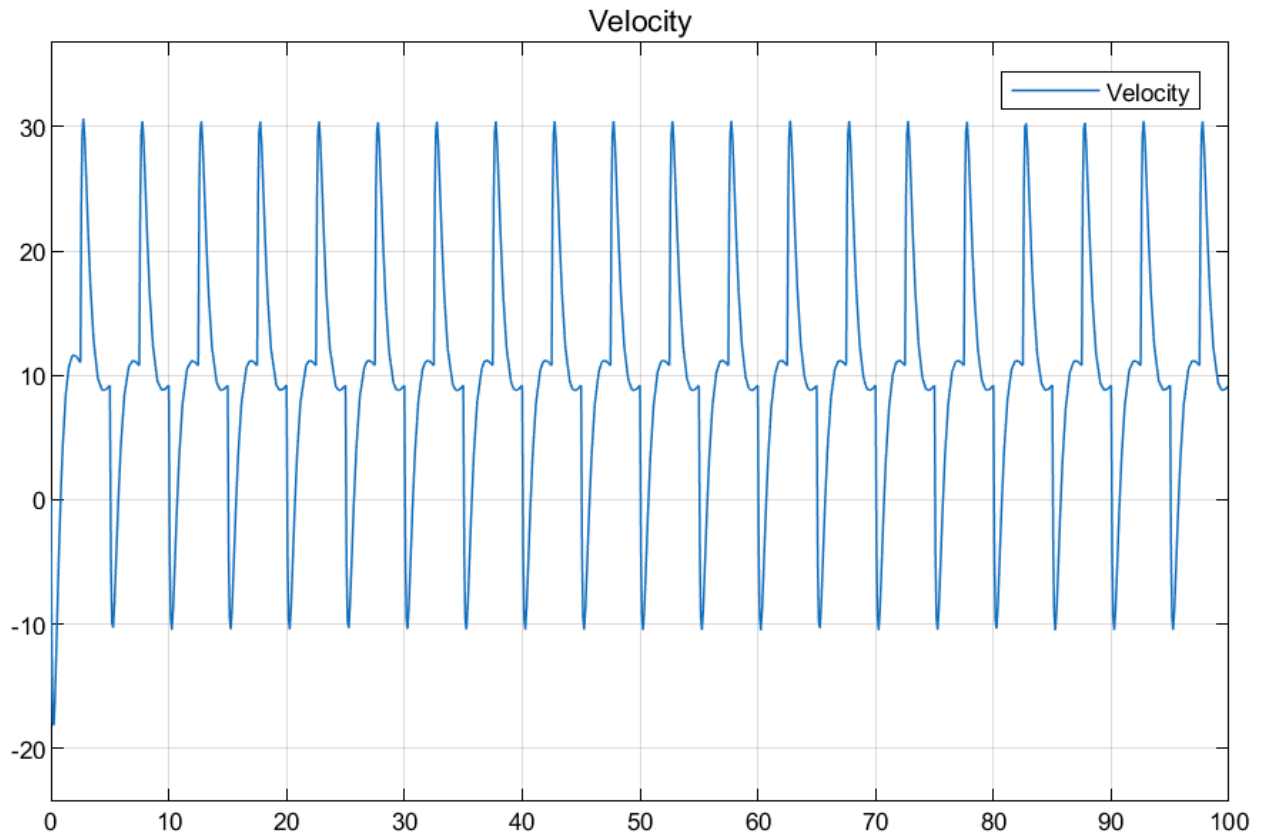
Plot. 9 [V2] Closed Loop Motor Velocity as a function of time (No Disturbance)



Plot. 10 [V2] Closed Loop Motor Velocity as a function of time (With Disturbance)



Plot. 11 [V3] Closed Loop Motor Velocity as a function of time (No Disturbance)



Plot. 12 [V3] Closed Loop Motor Velocity as a function of time (With Disturbance)

Motor Specifications:

Set	V1	V2	V3
Rise time	3.5911	1.4305	0.8772
Settling Time	6.3233	3.3390	2.6575
Overshoot	0	2.3746	5.0372
Steady-State Error	0	0.23	0.5

Table. 2 Motor specifications for Closed loop Motor velocity assuming no disturbance

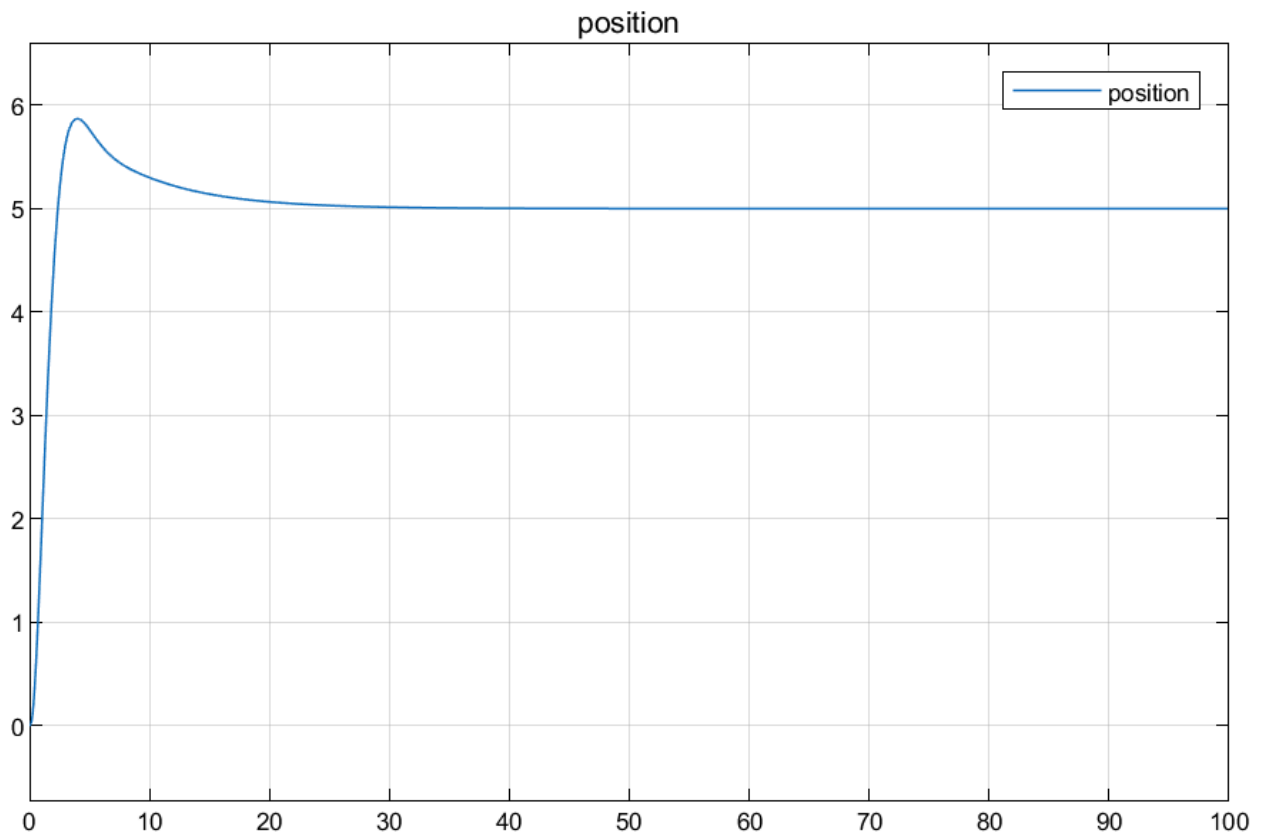
As the PID parameters increase in value, rise time decreases, settling time decreases, but overshoot and steady state error increase.

Input of 5 rad/s

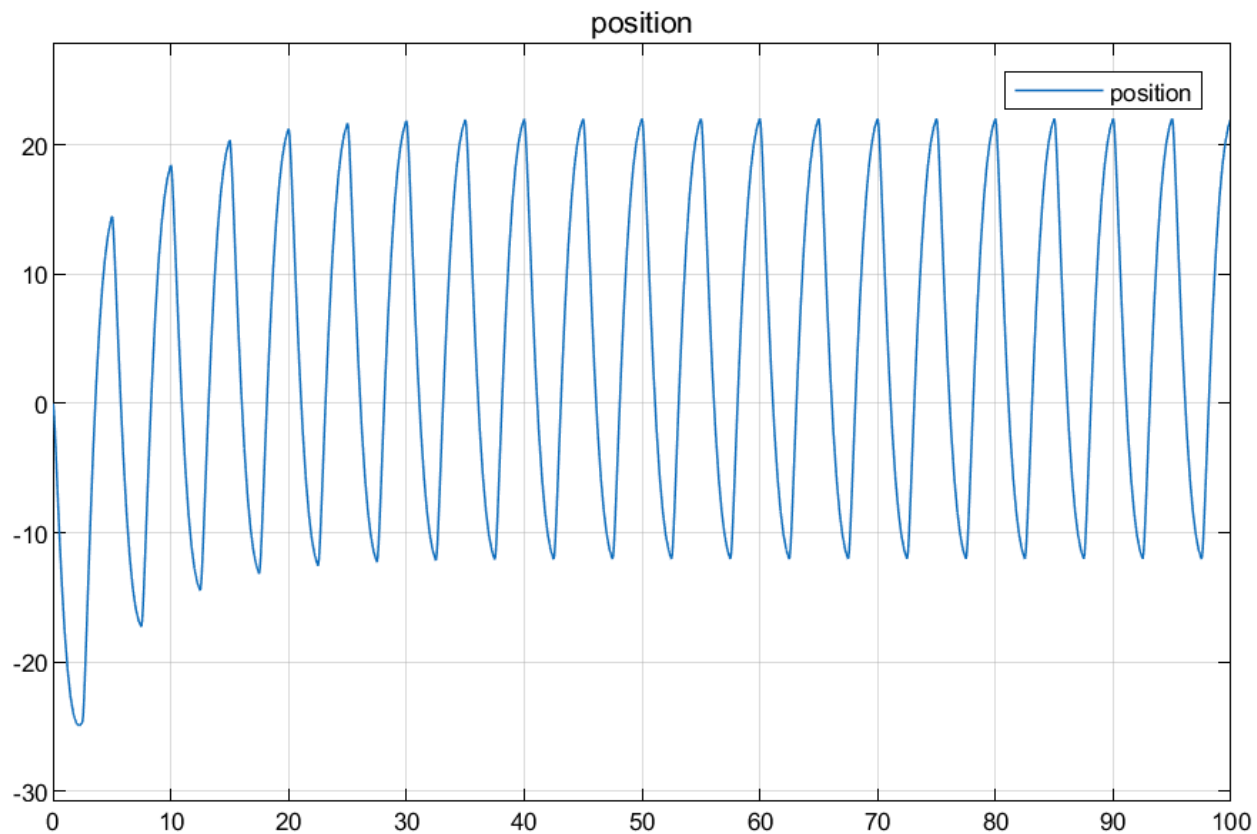
Parameters:

Set	Kp	Ki	Kd
P1	8.0	1.0	0.4
P2	15.0	3.0	0.8
P3	25.0	6.0	1.5

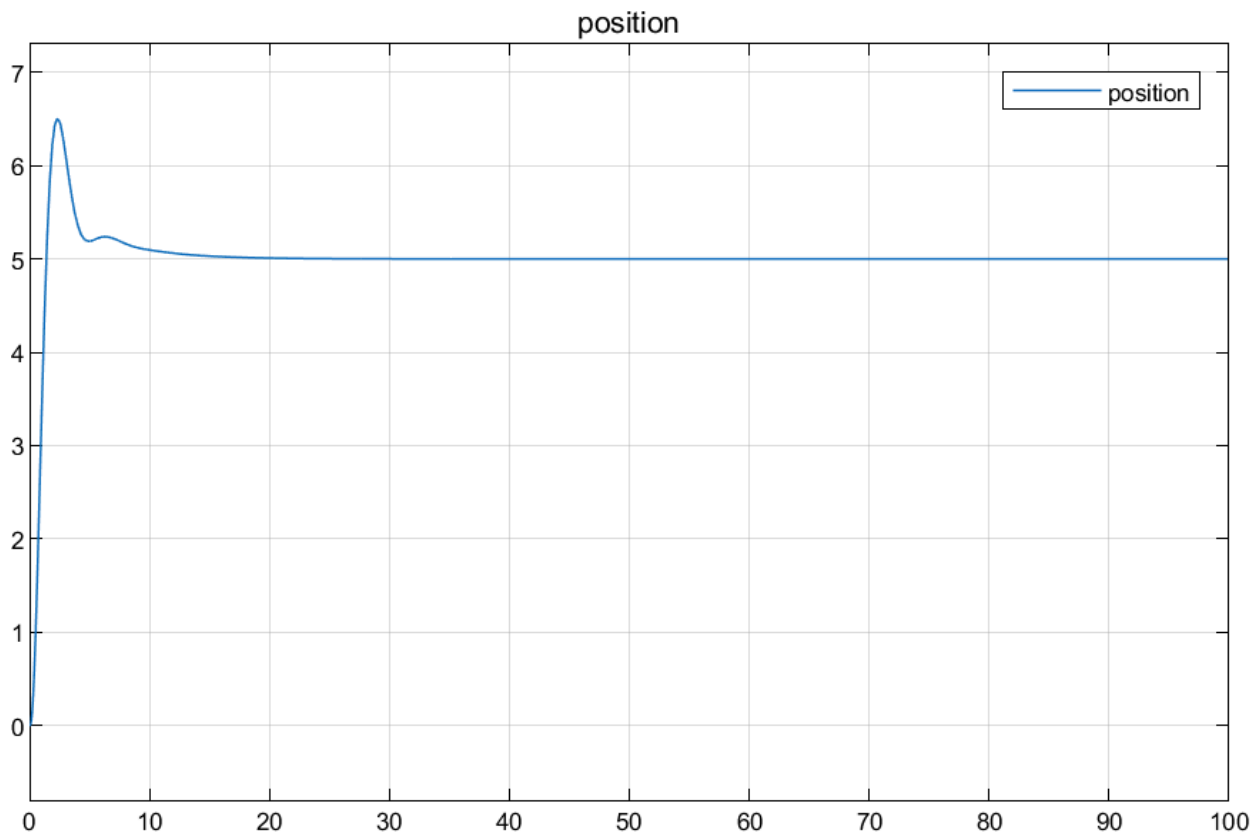
Table. 3 Input Parameters for closed loop motor of 5 rad/s input



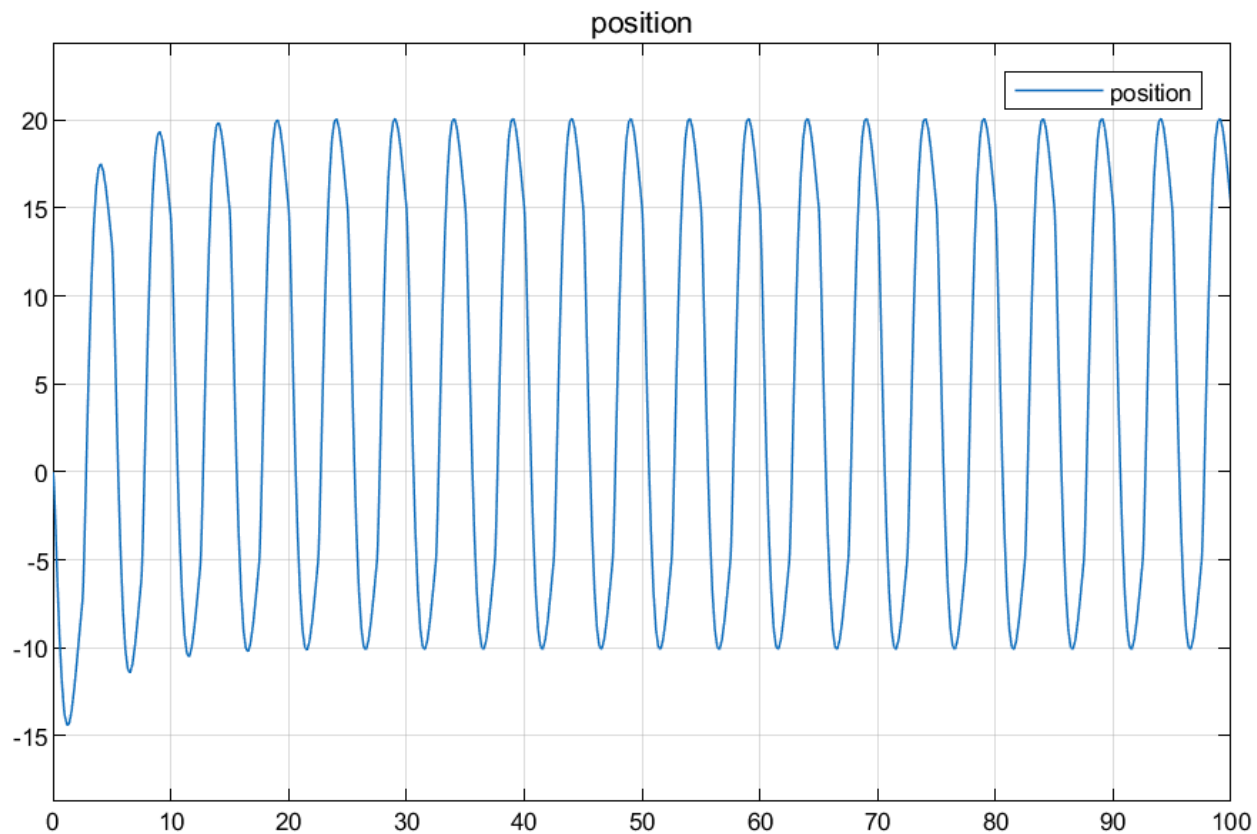
Plot. 13 [P1] Closed Loop Motor Position as a function of time (No Disturbance)



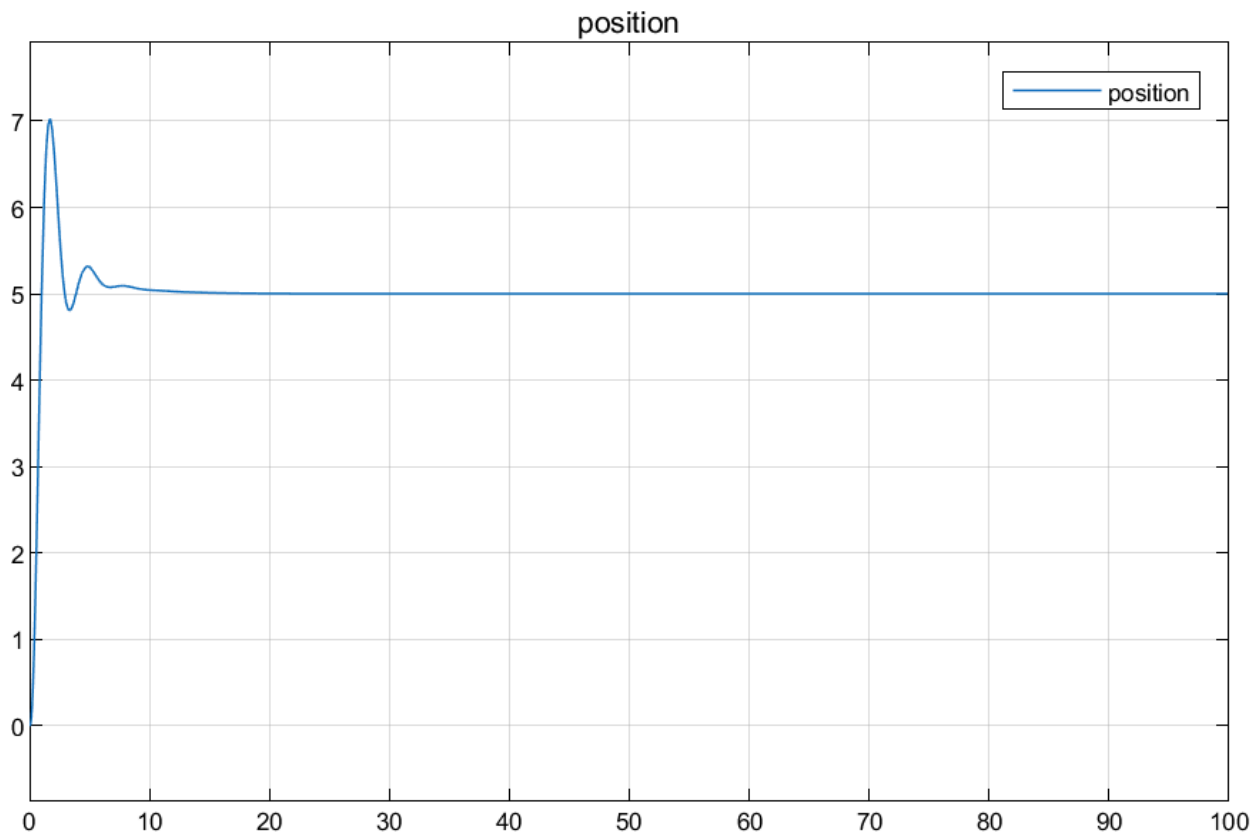
Plot. 14 [P1] Closed Loop Motor Position as a function of time (With Disturbance)



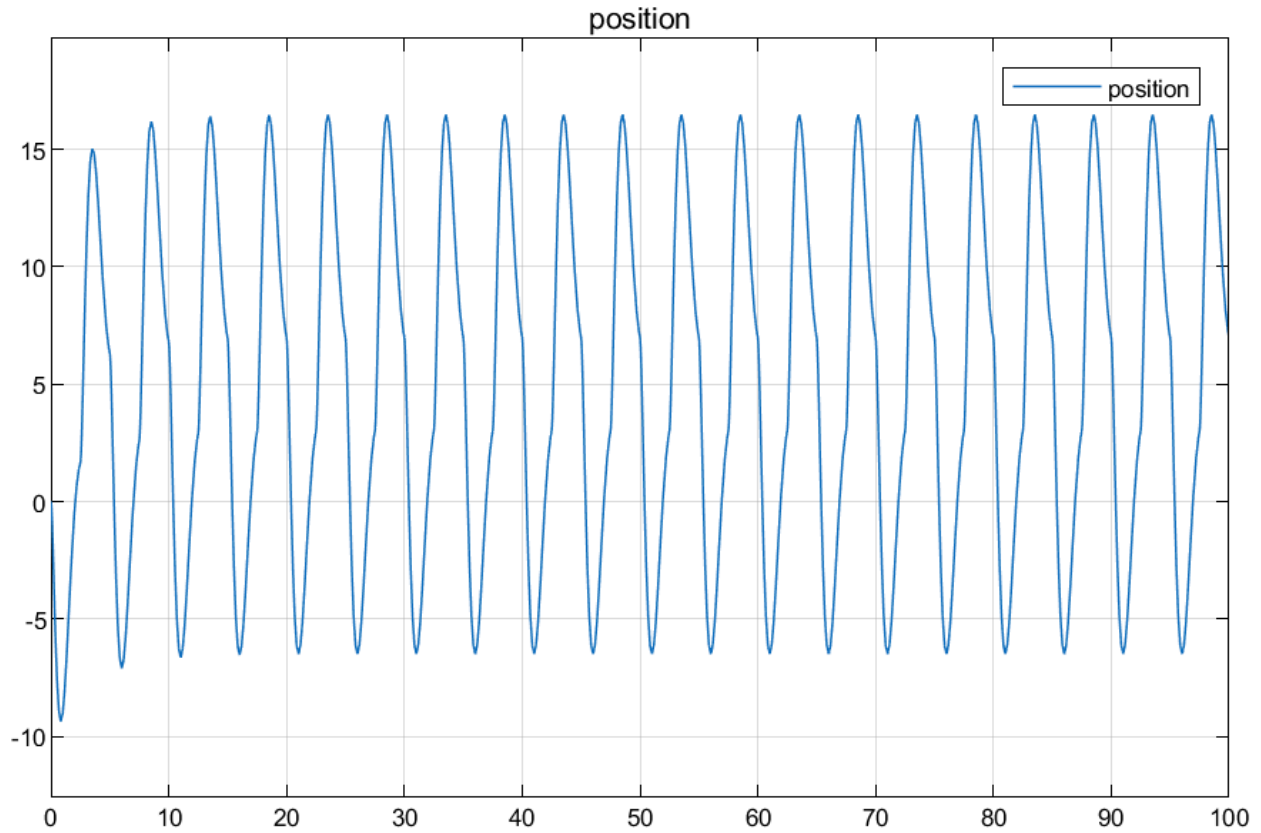
Plot. 15 [P2] Closed Loop Motor Position as a function of time (No Disturbance)



Plot. 16 [P2] Closed Loop Motor Position as a function of time (With Disturbance)



Plot. 17 [P3] Closed Loop Motor Position as a function of time (No Disturbance)



Plot. 18 [P3] Closed Loop Motor Position as a function of time (With Disturbance)

Motor Specifications:

Set	P1	P2	P3
Rise time	1.5595	0.9066	0.6322
Settling Time	17.2315	9.7927	6.1740
Overshoot	17.4311	30.0816	40.5810
Steady-State Error	0.8716	1.5	2.03

Table. 4 Motor specifications for Closed loop Motor position assuming no disturbance

As the PID parameters increase in value, rise time decreases, settling time decreases, but overshoot and steady state error increase.

8.0 Best Controller Gains

The results of this experiment clearly show the fundamental tradeoffs involved in tuning a PID controller for a DC motor system. Across both velocity and position control cases, increasing the controller gains led to faster system responses, but at the cost of increased overshoot and steady-state error.

For the velocity-controlled system (10 rad/s input), the three gain sets (V1, V2, V3) demonstrate a progression from conservative to aggressive tuning.

- V1 ($K_p=1.0$, $K_i=5.0$, $K_d=0.1$) produced no overshoot and zero steady-state error, but had the slowest rise time and settling time.
- V2 ($K_p=3.0$, $K_i=12.0$, $K_d=0.03$) provided a strong balance, significantly reducing rise time and settling time, while only introducing minimal overshoot and a very small steady-state error.
- V3 ($K_p=6.0$, $K_i=20.0$, $K_d=0.08$) achieved the fastest response, but at the cost of higher overshoot and larger steady-state error.

From this motor, V2 represents the best overall controller tuning for velocity control, as it balances speed and stability without introducing excessive oscillation or error.

For the position-controlled system (5 rad/s input):

- P1 ($K_p=8.0$, $K_i=1.0$, $K_d=0.4$) yields the lowest overshoot and smallest steady-state error, but has a relatively slow settling time.
- P2 ($K_p=15.0$, $K_i=3.0$, $K_d=0.8$) improves response speed but introduces significantly more overshoot and error.
- P3 ($K_p=25.0$, $K_i=6.0$, $K_d=1.5$) produces the fastest response but results in excessive overshoot and the largest steady-state error.

In this case, P1 is the most effective controller, as the large overshoot and error in P2 and P3 would likely be unacceptable in practical applications where position accuracy is critical.

9.0 Conclusion

In this experiment, open-loop and closed-loop control of a DC motor were modeled and compared using both transfer function and state-space representations. The open-loop system demonstrated stable behavior under ideal conditions but failed to compensate for disturbances, making it unreliable for real-world applications.

The implementation of a PID controller in the closed-loop system significantly improved performance by introducing feedback, allowing the system to track a desired reference and correct for disturbances. However, the tuning of the PID gains was shown to be critical in determining system performance.

It was observed that increasing controller gains reduces rise time and settling time, but also increases overshoot and steady-state error. Therefore, optimal controller design requires balancing these competing effects rather than maximizing any single performance metric.

For velocity control, the intermediate gain set (V2) provided the best balance of responsiveness and stability. For position control, the lowest gain set (P1) was most effective due to its reduced overshoot and improved accuracy. These results demonstrate that controller tuning must be tailored to the specific application and performance priorities.

In conclusion, closed-loop PID control offers substantial advantages over open-loop control, but careful tuning is essential to achieve a stable, accurate, and robust system.